

Wrap up

Zhichao Jiang

Sun Yat-sen University

Spring 2026

A big picture of causality

- Causal relationships are studied in many disciplines from different perspectives
- Our perspective: discover causal relationships from data
 - causal discovery: recover causal graphs from data
 - causal inference (this course): infer causal effect from data $\rightsquigarrow$ must consider potential outcomes
- The role of statistics in causal inference
 - identification: infer unobserved potential outcomes from observed data
 - statistical inference: properties of estimators, quantification of uncertainty

Causal inference

- Causal relationship = Correlation + Assumptions
 - we can only infer correlation from observed data
 - assumptions are crucial for connecting association with causation
 - plausibility of assumptions depends on design and domain knowledge
- Observed data = truth + spurious association + sample variation
 - statistical inference concerns sample variation
 - causal inference concerns both spurious association and sample variation

Acknowledgment

- Kosuke Imai, Harvard University, STAT186/GOV2002 Causal inference
- Peng Ding, UC Berkeley, STAT156/256 Causal inference
- James Heckman, U of Chicago, Econ 321 Empirical analysis

Study design

① Correlation $\rightarrow$ Causation

- Simpson's paradox
- potential outcomes $\rightsquigarrow$ causal effects
- fundamental problem: only one potential outcome is observed

② ACE in randomized experiments

- finite population framework
- Fisher randomization test for sharp null
- Neymanian inference: difference in means = simple regression with EHW variance
- covariate adjustment without modeling assumption
 $\rightsquigarrow$ linear regression with interaction between Z and X

③ No experiments $\rightsquigarrow$ observational study without unobserved confounders

- ignorability $\approx$ conditional randomized experiments
- various estimation methods: outcome regression, weighting, doubly robust, etc.

④ Difficulty of observational studies

- ignorability may fail due to latent confounding $\rightsquigarrow$ sensitivity analysis
- overlap assumption may fail $\rightsquigarrow$ sharp regression discontinuity design

⑤ Additional information (IV) to deal with latent confounding

- non-compliance problem: encouragement as IV $\rightsquigarrow$ CACE is identifiable
 - a special principal stratification problem: treatment receipt as post-treatment variable
 - various estimation methods under different assumptions
 - IV determined by forcing variable $\rightsquigarrow$ fuzzy regression discontinuity design
- IV with modeling assumptions
 - choice of IV relies on the knowledge of the underlying mechanism
 - two-stage least squares, etc.

Quantities of interest

- ① Overall ACE, ATT, and ACE in subgroups defined by pre-treatment variables X
 - different methods needed in different study designs
- ② ACE in subgroups defined by post-treatment variables S
 - cannot just condition on the observed S
 - principal stratification: condition on $(S(1), S(0))$
 - applications in non-compliance, truncation-by-death, surrogate evaluation problems
- ③ Treatment effect through a post-treatment variable (mediator) S
 - definition of NDE and NIE relies on nested potential outcomes
 - identification relies on cross-world independence assumption

Design trumps analysis

- Consider how you would conduct experiments even when you cannot
- Better design $\rightsquigarrow$ simpler causal inference
- More complex causal quantity $\rightsquigarrow$ more stringent assumptions

Important techniques

- Statistical inference and linear regression basics
- Derivations under finite population framework
 - simple random sampling
 - FRT, Neymanian inference
 - covariate adjustment
- Properties of conditional probabilities and expectations
 - different settings: unconfounded, confounded, IV, principal stratification, mediation
 - identification formulas and sensitivity analysis
 - estimation procedures under various modeling assumptions

Topics not covered in the course

- Heterogeneous treatment effects
 - conditional effect estimation
 - individual treatment rules
 - dynamic treatment regime
- Causal inference with panel data
 - time varying treatment
 - synthetic control method
- Causal inference without SUTVA
 - interference between units: spillover and diffusion effects
 - two-stage randomized experiments, network experiments
 - multiple versions of treatment